Expanded GS-PnP SSDU tuning

Dedicated preparation and experiment report

Current status: Prepared, not run.

1 Executive summary

The previous ground-truth-free GS-PnP selection was $\lambda = 0.1$, denoiser $\sigma = 0.075$ and initial $\tau_0 = 100/3$. Each was at the top of the previous finite search range. The five-image held-out measurement MSE was 0.0043654339. We prepare 27 new-grid configurations, including the old winning control and 26 previously untested combinations. Selecting among them must use only archived validation measurements, never image ground truth or the previously inspected 450-image test scores.

2 Tuning algorithm

Use the same five NPL-noisy validation images: CIGS 0733, Organic 0008, Perovskite 0283, Reference 0768 and Silicone 0589. For each image, retain the first 4,096 coarse rows of the 13,107 acquired Hadamard measurements in every reconstruction fold. Divide the remaining 9,011 acquired fine rows into the four previously frozen disjoint SSDU holdouts.

For candidate $\theta = (\lambda, \sigma, \tau_0)$, reconstruct from fit rows Θ_{ik} and score held-out rows Λ_{ik}:

$$L(\theta) = \frac{1}{5}\sum_{i=1}^{5}\frac{1}{4}\sum_{k=1}^{4}\frac{\|A_{\Lambda_{ik}}\widehat{x}_{\theta,ik} - y_{\Lambda_{ik}}\|_2^2}{|\Lambda_{ik}|}.$$

Keep the original normalised PCM operator and summed half-squared residual fidelity, without a fold-size rescaling. Use the same pretrained GS-DRUNet and paper-final z image.

Frozen parameter grid

Parameter	Candidate values
λ	0.10, 0.20, 0.40
Denoiser σ	0.075, 0.10, 0.15
Initial τ_0	33.3333, 50, 75

Freeze 100 maximum accepted steps, 20 line-search attempts, $\gamma = 0.1$, $\eta = 0.9$ and stopping tolerance 10^{-6}. Tune λ and τ_0 independently. Record accepted iterations, rejections, final τ, effective $\lambda\tau$, and time per reconstruction.

3 Step-by-step plan

1. Verify the frozen acquisition hashes, original four fold masks, solver equivalence and data-fidelity scale. Freeze source hashes and the candidate grid.
2. Reuse the original 20 fold scores of `gs_018` under control ID `gs_exp_001`, keeping original SHA256 provenance.
3. Score 26 new configurations across five images and four folds: 520 new GS reconstructions. Each can take up to 100 accepted iterations, with additional backtracking work.
4. Record all held-out losses and solver diagnostics. Exclude configurations with any failed fold. If more than nine candidates fail, withhold selection.
5. Select the minimum balanced SSDU loss, breaking ties within 10^{-10} by grid order. Freeze the candidate and every result hash.
6. Check whether the winning parameters are still at the expanded upper bounds. Only after freezing, consider a separately labelled retrospective image-quality comparison.

7. Refresh this report with actual scores and numerical diagnostics. Do not tune to test PSNR.

4 Experiment running

The expanded sweep has not begun. The intended GPU is tuni's RTX 4090, using an isolated CUDA UUID and deterministic cuBLAS/cuDNN. The original float64 noisy measurements and mask indices are reused, while neural solvers receive their required float32 values. The scoring code does not load targets. Case-level JSON files allow safe resumption. The old tuned-5 and tuned-50 selection records and the 450-image test benchmark are preserved.

5 Results and deliverables

Run status: 1 valid, 0 invalid, 26 untested/incomplete candidates. The expanded parameter sweep has not been completed or frozen. The old control's held-out MSE was 0.0043654339. No higher-value configuration has yet been selected. This is a preparation report, not a result claiming improved GS-PnP.

Candidate	λ	σ	τ_0	SSDU MSE	Status
gs_exp_001	0.10	0.075	33.3333	0.00436543	reused control
gs_exp_002	0.10	0.075	50.0000	–	not run
gs_exp_003	0.10	0.075	75.0000	–	not run
gs_exp_004	0.10	0.100	33.3333	–	not run
gs_exp_005	0.10	0.100	50.0000	–	not run
gs_exp_006	0.10	0.100	75.0000	–	not run
gs_exp_007	0.10	0.150	33.3333	–	not run
gs_exp_008	0.10	0.150	50.0000	–	not run
gs_exp_009	0.10	0.150	75.0000	–	not run
gs_exp_010	0.20	0.075	33.3333	–	not run
gs_exp_011	0.20	0.075	50.0000	–	not run
gs_exp_012	0.20	0.075	75.0000	–	not run
gs_exp_013	0.20	0.100	33.3333	–	not run
gs_exp_014	0.20	0.100	50.0000	–	not run
gs_exp_015	0.20	0.100	75.0000	–	not run
gs_exp_016	0.20	0.150	33.3333	–	not run
gs_exp_017	0.20	0.150	50.0000	–	not run
gs_exp_018	0.20	0.150	75.0000	–	not run
gs_exp_019	0.40	0.075	33.3333	–	not run
gs_exp_020	0.40	0.075	50.0000	–	not run
gs_exp_021	0.40	0.075	75.0000	–	not run
gs_exp_022	0.40	0.100	33.3333	–	not run
gs_exp_023	0.40	0.100	50.0000	–	not run
gs_exp_024	0.40	0.100	75.0000	–	not run
gs_exp_025	0.40	0.150	33.3333	–	not run
gs_exp_026	0.40	0.150	50.0000	–	not run
gs_exp_027	0.40	0.150	75.0000	–	not run

The dedicated folder holds the original grid, preflight, resumable numerical code, case-level results, frozen selection and an editable report in both TeX and PDF.

6 Conclusion

An expanded parameter grid can establish whether the original SSDU measurement-risk search was too narrow. It cannot by itself establish globally optimal reconstruction, or guarantee preservation of thin structures. Higher parameter values may worsen numerical stability. Correlated PCM noise and unmeasured Hadamard directions also limit the link between SSDU risk and image quality.

Primary papers. Yaman et al., *Self-Supervised Learning of Physics-Guided Reconstruction Neural Networks without Fully-Sampled Reference Data*, Magnetic Resonance in Medicine (2020), acquired measurement partitioning: `https://doi.org/10.1002/mrm.28378`.

Hurault, Leclaire and Papadakis, *Gradient Step Denoiser for Convergent Plug-and-Play*, ICLR (2022), gradient-step update and backtracking: `https://openreview.net/pdf?id=fPhKeld3Okz`.